INTERNATIONAL MONETARY FUND

CENTRAL AFRICAN REPUBLIC

April 13, 2020

REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

Approved By
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Prepared by the staffs of the International Monetary Fund (IMF) and the International Development Association (IDA)[1]

Central African Republic[1] Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	High[2]
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable
Application of judgement	No
Macroeconomic projections	This DSA update factors in the expected economic impact of the pandemic, with a significant decline in growth and trade flows and a sizeable widening of the fiscal and current account deficits in 2020. Growth is assumed to partially rebound in 2021 as the situation is assumed to start normalizing in the second half of 2020.
Financing strategy	Due to C.A.R.'s weak liquidity indicators, the country's strategy remains to seek additional grant financing from multilateral and bilateral donors to cover the residual financing gaps arising from the economic impact and mitigation efforts of the COVID-19 pandemic.
Realism tools flagged	n.a.
Mechanical risk rating under the external DSA	High
Mechanical risk rating under the public DSA.	High

[1] Debt coverage has not changed since the last DSA.
[2] With a score of 2.44, C.A.R.'s composite indicator, which is based on the October 2019 WEO and the 2018 CPIA, signals a weak debt-carrying capacity.

[1] This DSA has been prepared jointly by the IMF and World Bank, following the revised LIC-DSF Framework and Guidance Note (2017) in effect as of July 1, 2018.

This debt sustainability analysis (DSA) updates the joint World Bank-IMF analysis of December 2019 to reflect the most recent outlook, which has been substantially altered by the COVID-19 pandemic. The Central African Republic (C.A.R.) remains at high risk of external debt distress and overall high risk of debt distress, unchanged from the most recent DSA of December 2019.[2] These risks have increased owing to the high uncertainty surrounding the economic impact of the COVID-19 pandemic. While solvency indicators remain below their relevant thresholds, the disbursement under the RCF is projected to accentuate the breaches of the liquidity indicators over the medium term. Sensitivity of debt indicators to standard stress tests, the high uncertainty surrounding macroeconomic projections, a volatile security environment, and sizeable contingent liabilities are all considerations supporting the high-risk assessment. Debt is projected to remain sustainable over the medium term provided that the authorities move ahead with the policies and structural reforms committed under the ECF arrangement once the effects of the COVID crisis wear off. Given the difficult debt situation, staffs recommend that the government's investment program requires grant financing, with concessional debt financing to be considered in exceptional cases.

The current macroeconomic framework reflects currently available information. Updates of the economic impact and policy response to the COVID-19 crisis are rapidly evolving and risks are heavily tilted to the downside.

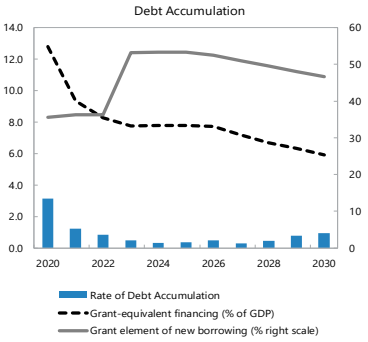
[2] Country Report No. 20/1, January 2020.

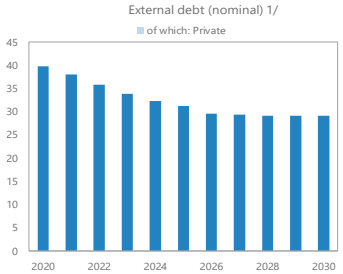
Table 1. Central African Republic: External Debt Sustainability Framework, Baseline Scenario, 2017–40

(Percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**35.4**	**37.2**	**37.2**	**39.7**	**38.0**	**35.7**	**33.8**	**32.3**	**31.1**	**29.1**	**27.4**	**28.7**	**32.4**
of which: public and publicly guaranteed (PPG)	**35.4**	**37.2**	**37.2**	**39.7**	**38.0**	**35.7**	**33.8**	**32.3**	**31.1**	**29.1**	**27.4**	**28.7**	**32.4**
Change in external debt	0.4	1.8	0.0	2.5	-1.7	-2.2	-1.9	-1.5	-1.1	0.1	-0.1		
Identified net debt-creating flows	**...**	**3.9**	**3.8**	**4.9**	**2.7**	**2.1**	**2.6**	**2.7**	**2.7**	**0.2**	**-0.2**	**3.9**	**1.7**
Non-interest current account deficit	**7.7**	**7.8**	**4.8**	**5.5**	**5.1**	**5.1**	**5.6**	**5.6**	**5.5**	**2.8**	**2.7**	**7.1**	**4.2**
Deficit in balance of goods and services	16.4	17.8	18.3	17.3	15.9	15.0	14.5	14.2	14.0	8.8	8.5	**15.0**	**12.3**
Exports	15.9	15.9	15.7	14.4	14.7	15.0	15.0	15.0	15.1	15.2	15.5		
Imports	32.4	33.7	34.1	31.7	30.6	30.0	29.5	29.3	29.1	24.0	24.0		
Net current transfers (negative = inflow)	-8.8	-9.9	-13.6	-11.8	-10.8	-9.9	-8.9	-8.5	-8.3	-7.3	-5.8	**-7.8**	**-8.7**
of which: official	-3.2	-4.4	-7.3	-6.9	-5.5	-4.8	-4.0	-3.8	-3.7	-3.9	-2.9		
Other current account flows (negative = net inflow)	0.0	-0.1	0.0	0.0	0.0	-0.1	-0.1	-0.1	-0.2	1.3	0.0	**-0.1**	**0.6**
Net FDI (negative = inflow)	**-0.8**	**-0.8**	**-1.1**	**-0.4**	**-1.0**	**-1.4**	**-1.5**	**-1.5**	**-1.5**	**-1.8**	**-2.3**	**-1.3**	**-1.5**
Endogenous debt dynamics 2/	**...**	**-3.1**	**0.2**	**-0.2**	**-1.3**	**-1.6**	**-1.5**	**-1.4**	**-1.3**	**-0.7**	**-0.6**		
Contribution from nominal interest rate	0.1	0.2	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.3		
Contribution from real GDP growth	-1.4	-1.2	-1.1	-0.4	-1.5	-1.8	-1.7	-1.6	-1.5	-0.9	-0.9		
Contribution from price and exchange rate changes	...	-2.0	1.1	...	...	...	...	...	...	...	...		
Residual 3/	**...**	**-2.2**	**-3.8**	**-2.4**	**-4.4**	**-4.3**	**-4.5**	**-4.3**	**-3.8**	**-0.2**	**0.1**	**-3.0**	**-2.4**
of which: exceptional financing	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**18.6**	**21.1**	**20.8**	**20.0**	**19.0**	**17.9**	**16.9**	**14.5**	**17.0**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**118.3**	**147.2**	**141.2**	**133.4**	**126.7**	**119.0**	**112.2**	**95.4**	**109.7**		
PPG debt service-to-exports ratio	**1.8**	**3.8**	**3.6**	**7.2**	**6.9**	**8.4**	**9.1**	**11.5**	**12.3**	**8.0**	**6.3**		
PPG debt service-to-revenue ratio	**3.7**	**6.8**	**6.6**	**12.1**	**9.8**	**11.9**	**12.7**	**15.8**	**16.7**	**10.1**	**7.0**		
Gross external financing need (Million of U.S. dollars)	148.9	173.3	96.3	143.2	127.7	134.4	161.2	186.1	199.6	104.8	116.5		
Key macroeconomic assumptions													
Real GDP growth (in percent)	4.5	3.8	3.0	1.0	4.0	5.0	5.0	5.0	5.0	3.4	3.4	**-0.2**	**4.1**
GDP deflator in US dollar terms (change in percent)	8.6	6.0	-2.9	1.8	3.5	3.0	2.8	2.7	2.7	2.5	2.5	**2.5**	**2.9**
Effective interest rate (percent) 4/	0.4	0.5	0.4	0.4	0.4	0.4	0.5	0.6	0.6	0.8	1.2	**1.1**	**0.6**
Growth of exports of G&S (US dollar terms, in percent)	0.3	10.2	-1.3	-6.1	10.3	10.2	7.9	8.3	8.2	6.2	6.2	**6.1**	**6.9**
Growth of imports of G&S (US dollar terms, in percent)	6.8	14.6	1.0	-4.3	4.2	5.8	6.1	7.0	7.2	6.0	6.0	**7.8**	**3.9**
Grant element of new public sector borrowing (in percent)	...	...	...	35.6	36.3	36.3	53.1	53.3	53.2	46.6	31.9	**...**	**46.8**
Government revenues (excluding grants, in percent of GDP)	7.8	8.9	8.7	8.5	10.3	10.6	10.7	10.9	11.1	12.1	14.0	**8.0**	**11.0**
Aid flows (in Million of US dollars) 5/	14434.2	24576.8	9719.7	283.5	230.0	220.0	210.3	217.5	230.9	226.3	164.9		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	12.8	9.3	8.3	7.8	7.8	7.8	5.9	2.6	**...**	**8.0**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	80.1	88.9	88.4	89.7	87.2	85.3	80.4	64.2	**...**	**84.1**
Nominal GDP (Million of US dollars)	2,072	2,280	2,279	2,345	2,525	2,732	2,948	3,178	3,427	4,858	8,687		
Nominal dollar GDP growth	13.5	10.1	0.0	2.9	7.7	8.2	7.9	7.8	7.8	6.0	6.0	**2.1**	**7.1**
Memorandum items:													
PV of external debt 7/	...	...	18.6	21.1	20.8	20.0	19.0	17.9	16.9	14.5	17.0		
In percent of exports	...	...	118.3	147.2	141.2	133.4	126.7	119.0	112.2	95.4	109.9		
Total external debt service-to-exports ratio	1.8	3.8	3.6	7.2	6.9	8.4	9.1	11.5	12.3	8.0	6.3		
PV of PPG external debt (in Million of US dollars)			424.5	495.9	524.7	545.9	559.4	569.0	580.1	704.7	1477.5		
(PVt-PVt-1)/GDPt-1 (in percent)				3.1	1.2	0.8	0.5	0.3	0.3	1.0	1.3		
Non-interest current account deficit that stabilizes debt ratio	7.3	6.0	4.8	3.0	6.8	7.3	7.5	7.2	6.6	2.7	2.8		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes valuation adjustments. For projections also includes contribution from price and exchange rate changes. High value of the residual is related to capital grants which are not captured in this presentation.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

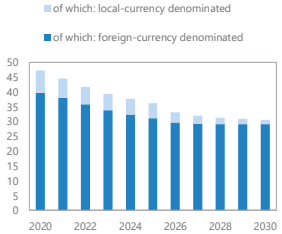
Table 2. Central African Republic: Public Sector Debt Sustainability Framework, Baseline Scenario 2017–40

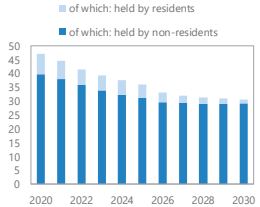
(Percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**50.3**	**50.0**	**47.8**	**47.1**	**44.5**	**41.5**	**39.3**	**37.5**	**36.1**	**30.6**	**28.7**	**44.7**	**36.7**
of which: external debt	**35.4**	**37.2**	**37.2**	**39.7**	**38.0**	**35.7**	**33.8**	**32.3**	**31.1**	**29.1**	**27.4**	**28.7**	**32.4**
Change in public sector debt	-3.6	-0.2	-2.2	-0.7	-2.6	-3.0	-2.3	-1.7	-1.5	-0.3	-0.1		
Identified debt-creating flows	**-7.9**	**0.0**	**-2.8**	**-0.8**	**-2.7**	**-3.0**	**-2.1**	**-1.7**	**-1.2**	**-0.3**	**0.0**	**0.8**	**-1.5**
Primary deficit	**0.7**	**0.6**	**-1.8**	**1.8**	**-0.7**	**0.1**	**0.9**	**1.2**	**1.4**	**1.1**	**1.2**	**0.8**	**0.8**
Revenue and grants	12.8	16.6	18.3	19.6	18.9	18.3	17.5	17.4	17.3	16.7	15.9	**13.9**	**17.6**
of which: grants	5.0	7.8	9.6	11.0	8.7	7.6	6.8	6.5	6.3	4.7	1.9		
Primary (noninterest) expenditure	13.5	17.2	16.6	21.3	18.3	18.4	18.4	18.6	18.7	17.8	17.1	**14.6**	**18.4**
Automatic debt dynamics	**-8.6**	**-0.8**	**-1.4**	**-1.5**	**-2.8**	**-3.0**	**-2.7**	**-2.5**	**-2.3**	**-1.4**	**-1.2**		
Contribution from interest rate/growth differential	-5.2	-2.2	-2.3	-1.2	-2.6	-2.9	-2.6	-2.4	-2.3	-1.5	-1.2		
of which: contribution from average real interest rate	-2.8	-0.3	-0.8	-0.7	-0.8	-0.7	-0.6	-0.5	-0.5	-0.4	-0.3		
of which: contribution from real GDP growth	-2.3	-1.9	-1.4	-0.5	-1.8	-2.1	-2.0	-1.9	-1.8	-1.0	-0.9		
Contribution from real exchange rate depreciation	-3.5	1.3	0.9	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.2**	**0.4**	**-1.0**	**0.8**	**-0.1**	**-0.3**	**-0.4**	**-0.3**	**0.0**	**0.0**	**0.1**	**-0.1**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Change in deposits	...	0.2	0.4	-0.8	0.8	-0.1	-0.3	-0.4	-0.3	0.0	0.0		
Residual	**4.3**	**-0.2**	**0.6**	**-0.4**	**-0.2**	**-0.2**	**-0.2**	**-0.1**	**-0.3**	**0.0**	**0.0**	**2.0**	**-0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**29.4**	**28.5**	**27.3**	**25.7**	**24.4**	**23.2**	**21.8**	**15.9**	**18.3**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**160.4**	**145.6**	**144.1**	**141.0**	**139.6**	**133.2**	**126.0**	**95.4**	**115.1**		
Debt service-to-revenue and grants ratio 3/	**9.2**	**18.6**	**17.1**	**20.9**	**10.1**	**12.2**	**14.5**	**19.5**	**23.1**	**13.6**	**9.3**		
Gross financing need 4/	1.7	3.4	1.4	5.7	1.2	2.3	3.4	4.5	5.3	3.4	2.6		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	4.5	3.8	3.0	1.0	4.0	5.0	5.0	5.0	5.0	3.4	3.4	**-0.2**	**4.1**
Average nominal interest rate on external debt (in percent)	0.5	0.5	0.4	0.4	0.4	0.4	0.5	0.6	0.6	0.8	1.2	**1.1**	**0.6**
Average real interest rate on domestic debt (in percent)	-5.3	-0.4	-0.7	-0.4	-0.1	0.0	0.3	0.9	1.4	3.3	5.7	**-1.9**	**1.1**
Real exchange rate depreciation (in percent, + indicates depreciation)	-10.9	4.0	2.4	...	...	...	...	...	...	...	...	**2.9**	**...**
Inflation rate (GDP deflator, in percent)	6.4	1.3	2.4	2.3	2.5	2.5	2.5	2.5	2.5	2.5	2.5	**4.5**	**2.7**
Growth of real primary spending (deflated by GDP deflator, in percent)	22.4	31.8	-0.9	30.1	-10.9	5.7	5.1	5.9	5.8	3.0	2.9	**3.1**	**5.2**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	4.3	0.8	0.5	2.5	1.9	3.1	3.2	2.9	2.8	1.5	1.2	**1.9**	**2.3**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments plus social security, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

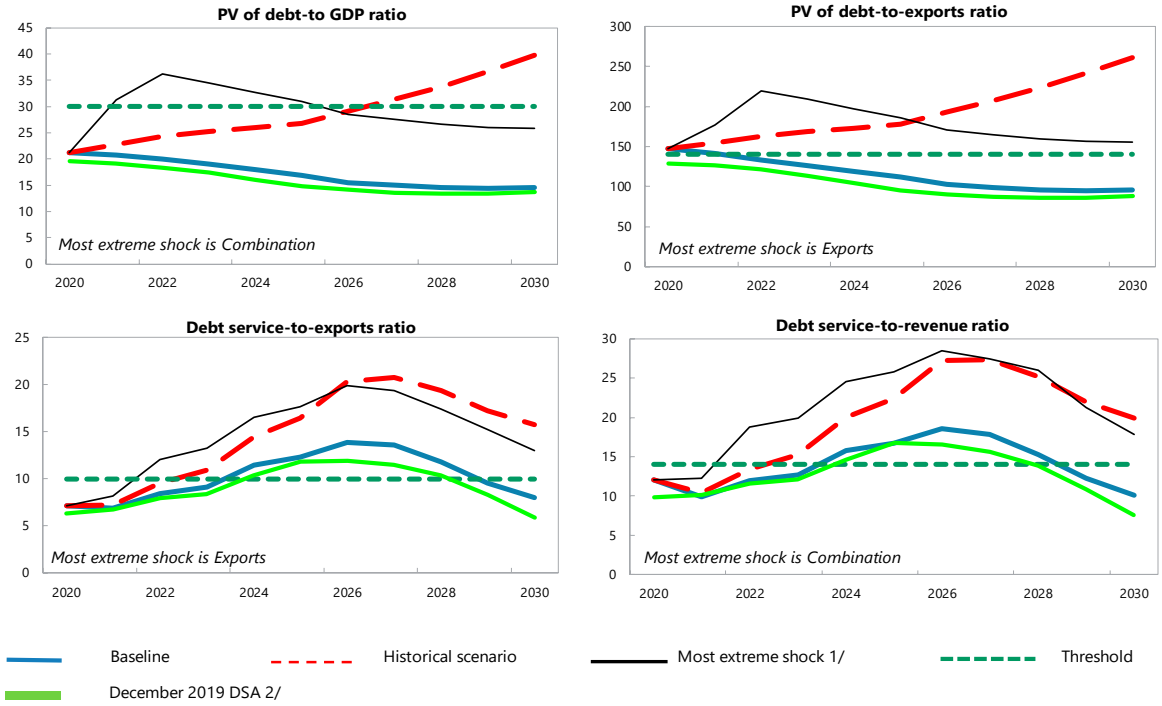
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Central African Republic: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2020–30



Customization of Default Settings	Size	Interactions
Standardized Tests	Yes	
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices 3/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

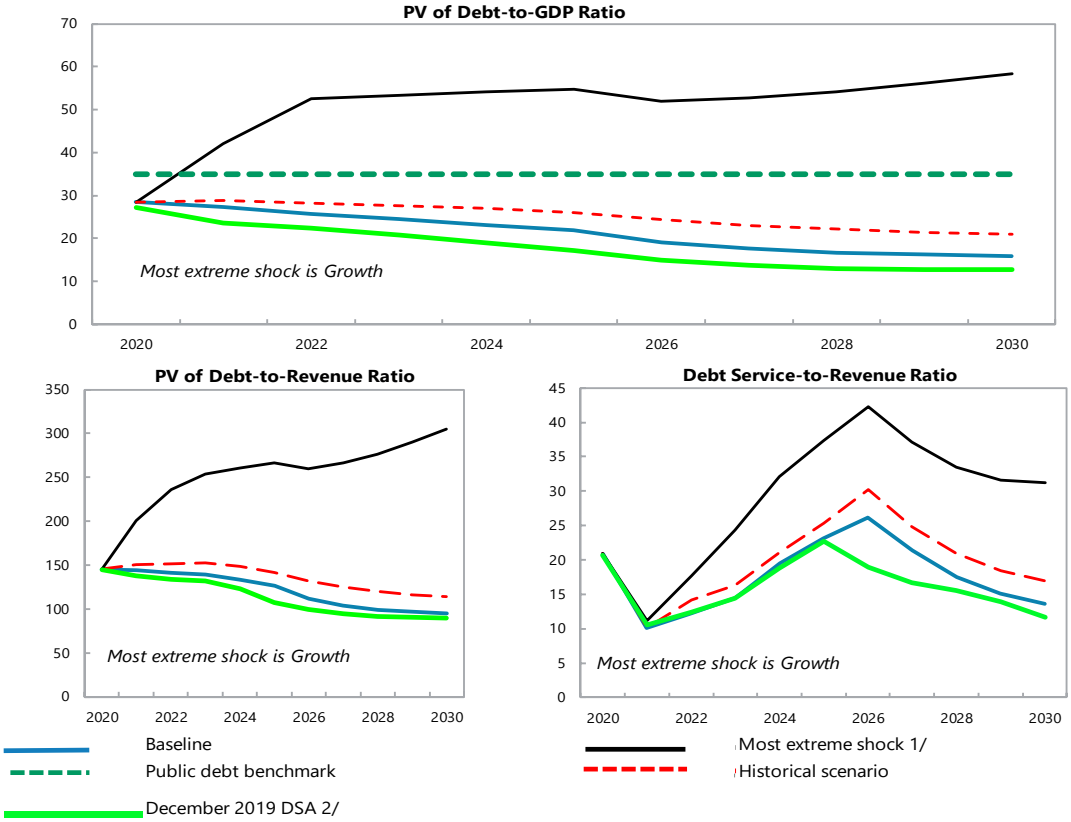
Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	0.8%	0.8%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	30	30
Avg. grace period	6	6

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.
2/ The alternative scenario corresponds to the latest DSA (December 2019)
3/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Central African Republic: Indicators of Public Debt Under Alternative Scenarios, 2020–30



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	77%	77%
Domestic medium and long-term	8%	8%
Domestic short-term	15%	15%
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	0.8%	0.8%
Avg. maturity (incl. grace period)	30	30
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	5.5%	5.5%
Avg. maturity (incl. grace period)	2	2
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	2%	2.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The alternative scenario corresponds to the latest DSA (December 2019)

Table 3. Central African Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020–30

	Projections										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	**21.1**	**20.8**	**20.0**	**19.0**	**17.9**	**16.9**	**15.5**	**14.9**	**14.5**	**14.4**	**14.5**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	21.1	22.8	24.3	25.3	26.0	26.8	29.1	31.3	33.8	36.6	39.7
A2. Alternative Scenario : December 2019 DSA	19.6	19.1	18.3	17.3	15.9	14.8	14.1	13.5	13.3	13.4	13.7
B. Bound Tests											
B1. Real GDP growth	21.1	25.6	30.3	28.8	27.1	25.7	23.5	22.6	22.0	21.9	22.0
B2. Primary balance	21.1	22.5	23.1	22.2	21.0	19.9	18.3	17.7	17.2	17.0	17.0
B3. Exports	21.1	22.1	23.7	22.6	21.4	20.3	18.6	18.0	17.5	17.2	17.1
B4. Other flows 2/	21.1	24.8	27.4	26.1	24.8	23.6	21.7	21.0	20.4	19.8	19.5
B5. One-time 30 percent nominal depreciation	21.1	26.1	21.6	20.4	19.2	18.1	16.5	15.8	15.4	15.5	15.8
B6. Combination of B1-B5	21.1	31.2	36.2	34.5	32.7	31.0	28.5	27.5	26.5	26.0	25.8
C. Tailored Tests											
C1. Combined contingent liabilities	21.1	27.1	27.0	26.2	24.9	23.7	21.8	21.1	20.7	20.5	20.5
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**
PV of debt-to-exports ratio											
Baseline	**147.2**	**141.2**	**133.4**	**126.7**	**119.0**	**112.2**	**102.8**	**98.7**	**95.8**	**95.1**	**95.4**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	147.2	154.9	162.3	168.6	172.8	177.7	192.9	207.3	223.0	241.3	261.2
A2. Alternative Scenario : December 2019 DSA	128.8	126.8	121.6	114.0	104.8	94.9	90.6	87.1	85.7	86.0	88.0
B. Bound Tests											
B1. Real GDP growth	147.2	141.2	133.4	126.7	119.0	112.2	102.8	98.7	95.8	95.1	95.4
B2. Primary balance	147.2	153.3	153.9	148.1	139.8	131.8	121.1	117.0	113.7	112.3	112.0
B3. Exports	147.2	177.2	219.3	208.9	196.9	186.2	171.0	164.9	159.9	156.9	155.6
B4. Other flows 2/	147.2	168.5	182.7	174.4	165.0	156.5	144.0	139.3	134.4	130.7	128.3
B5. One-time 30 percent nominal depreciation	147.2	141.2	114.9	108.8	101.8	95.5	87.3	83.5	80.9	81.3	82.7
B6. Combination of B1-B5	147.2	188.1	160.5	187.4	176.8	167.3	153.7	148.3	142.7	139.8	138.4
C. Tailored Tests											
C1. Combined contingent liabilities	147.2	183.9	179.9	175.0	165.7	156.9	144.5	139.9	136.5	135.2	135.1
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**
Debt service-to-exports ratio											
Baseline	**7.2**	**6.9**	**8.4**	**9.1**	**11.5**	**12.3**	**13.9**	**13.6**	**11.8**	**9.6**	**8.0**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	7.2	7.2	9.5	10.9	14.5	16.4	20.3	20.8	19.4	17.2	15.7
A2. Alternative Scenario : December 2019 DSA	6.3	6.7	8.0	8.4	10.3	11.9	11.9	11.4	10.4	8.3	**5.9**
B. Bound Tests											
B1. Real GDP growth	7.2	6.9	8.4	9.1	11.5	12.3	13.9	13.6	11.8	9.6	8.0
B2. Primary balance	7.2	6.8	8.6	9.4	11.7	12.5	14.1	13.7	12.4	10.6	9.1
B3. Exports	7.2	8.2	12.0	13.2	16.5	17.6	19.9	19.4	17.4	15.2	13.0
B4. Other flows 2/	7.2	6.9	8.8	9.8	12.1	12.9	14.4	14.1	13.5	12.3	10.6
B5. One-time 30 percent nominal depreciation	7.2	6.9	8.4	8.9	11.2	12.1	13.7	13.4	11.6	8.5	7.0
B6. Combination of B1-B5	7.2	7.6	10.8	11.6	14.4	15.4	17.4	17.0	16.3	13.5	11.5
C. Tailored Tests											
C1. Combined contingent liabilities	7.2	6.8	9.0	9.7	12.1	12.8	14.4	14.0	12.2	10.0	8.4
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**	**10.0**
Debt service-to-revenue ratio											
Baseline	**12.1**	**9.8**	**11.9**	**12.7**	**15.8**	**16.7**	**18.6**	**17.9**	**15.3**	**12.2**	**10.1**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	12.1	10.4	13.4	15.2	20.0	22.3	27.2	27.4	25.1	22.0	19.8
A2. Alternative Scenario : December 2019 DSA	9.9	10.1	11.6	12.1	14.6	16.8	16.5	15.6	13.9	10.8	**7.6**
B. Bound Tests	9.9	10.1	11.6	12.1	14.6	16.8	16.5	15.6	13.9	10.8	**7.6**
B1. Real GDP growth	12.1	12.1	18.0	19.3	24.0	25.4	28.1	27.1	23.1	18.5	15.3
B2. Primary balance	12.1	9.8	12.1	13.1	16.2	17.1	18.8	18.0	16.1	13.6	11.5
B3. Exports	12.1	9.9	12.2	13.3	16.5	17.4	19.2	18.4	16.3	14.0	11.8
B4. Other flows 2/	12.1	9.8	12.4	13.6	16.7	17.5	19.3	18.5	17.5	15.7	13.4
B5. One-time 30 percent nominal depreciation	12.1	12.3	14.9	15.5	19.4	20.6	22.9	22.1	18.9	13.7	11.1
B6. Combination of B1-B5	12.1	12.3	18.7	19.9	24.5	25.8	28.5	27.4	26.0	21.2	17.9
C. Tailored Tests											
C1. Combined contingent liabilities	12.1	9.8	12.7	13.5	16.6	17.5	19.2	18.5	15.9	12.8	10.6
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	**14.0**	**14.0**	**14.0**	**14.0**	**14.0**	**14.0**	**14.0**	**14.0**	**14.0**	**14.0**	**14.0**

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Includes official and private transfers and FDI.

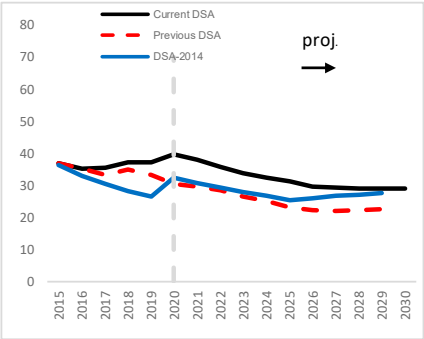
Table 4. Central African Republic: Sensitivity Analysis for Key Indicators of Public Debt, 2020-30

	Projections										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	**28.5**	**27.3**	**25.7**	**24.4**	**23.2**	**21.8**	**19.0**	**17.6**	**16.7**	**16.3**	**15.9**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	28.5	28.8	28.3	27.6	26.9	25.9	24.4	23.1	22.1	21.4	20.9
A2. Alternative Scenario : December 2019 DSA	27.1	23.6	22.4	20.8	18.9	17.1	14.9	13.8	13.0	12.8	12.8
B. Bound Tests											
B1. Real GDP growth	28.5	42.1	52.5	53.3	54.1	54.7	52.0	52.7	54.1	56.2	58.4
B2. Primary balance	28.5	29.8	29.8	28.2	26.8	25.4	22.2	20.8	19.8	19.2	18.8
B3. Exports	28.5	28.5	29.1	27.7	26.3	24.9	21.8	20.4	19.4	18.8	18.3
B4. Other flows 2/	28.5	31.3	33.1	31.5	30.1	28.5	25.2	23.7	22.6	21.7	20.9
B5. One-time 30 percent nominal depreciation	28.5	40.1	37.4	35.0	32.5	29.8	25.0	22.3	20.2	18.8	17.6
B6. Combination of B1-B5	28.5	32.3	32.3	31.6	30.7	29.6	26.4	24.8	23.8	23.2	22.8
C. Tailored Tests											
C1. Combined contingent liabilities	28.5	37.0	34.3	32.3	30.8	29.2	25.8	24.3	23.3	22.7	22.2
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	**35.0**	**35.0**	**35.0**	**35.0**	**35.0**	**35.0**	**35.0**	**35.0**	**35.0**	**35.0**	**35.0**
PV of Debt-to-Revenue Ratio											
Baseline	**145.6**	**144.1**	**141.0**	**139.6**	**133.2**	**126.0**	**111.2**	**103.7**	**99.0**	**96.7**	**95.4**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	146	150	151	153	148	141	132	125	120	117	114
A2. Alternative Scenario : December 2019 DSA	20.7	10.5	12.4	14.4	18.9	22.7	18.9	16.7	15.5	13.9	11.7
B. Bound Tests											
B1. Real GDP growth	145.6	200.9	236.2	253.9	260.9	266.0	259.7	266.2	276.6	290.2	305.2
B2. Primary balance	145.6	157.6	163.4	161.1	154.2	146.5	130.5	122.5	117.3	114.3	112.2
B3. Exports	145.6	150.4	159.5	158.3	151.4	143.6	127.9	120.2	115.0	111.6	109.2
B4. Other flows 2/	145.6	165.3	181.4	180.4	173.0	164.5	147.7	139.7	133.6	128.9	125.2
B5. One-time 30 percent nominal depreciation	145.6	219.9	212.0	206.7	193.0	177.2	150.6	134.8	122.9	114.5	107.9
B6. Combination of B1-B5	145.6	165.5	163.4	168.0	164.4	159.0	144.9	137.2	132.6	130.4	129.2
C. Tailored Tests											
C1. Combined contingent liabilities	145.6	195.4	188.0	184.7	177.1	168.7	151.4	143.1	137.9	135.0	133.1
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	**20.9**	**10.1**	**12.2**	**14.5**	**19.5**	**23.1**	**26.2**	**21.4**	**17.5**	**15.0**	**13.6**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2040 1/	20.9	10.1	14.2	16.3	21.0	25.4	30.2	24.9	20.9	18.4	16.9
A2. Alternative Scenario : December 2019 DSA	20.7	10.5	12.4	14.4	18.9	22.7	18.9	16.7	15.5	13.9	11.7
B. Bound Tests											
B1. Real GDP growth	20.9	11.1	17.7	24.4	32.1	37.4	42.3	37.1	33.5	31.6	31.2
B2. Primary balance	20.9	10.0	15.8	19.6	21.8	23.7	26.3	21.5	18.2	16.2	14.9
B3. Exports	20.9	10.1	12.3	14.7	19.8	23.3	26.4	21.6	18.0	16.2	14.7
B4. Other flows 2/	20.9	10.1	12.5	15.0	20.1	23.6	26.6	21.8	19.0	17.5	16.0
B5. One-time 30 percent nominal depreciation	20.9	10.6	14.2	16.6	22.4	26.2	29.7	24.9	20.6	17.4	15.4
B6. Combination of B1-B5	20.9	10.4	13.1	17.4	24.1	28.6	32.4	27.5	23.6	21.0	19.6
C. Tailored Tests											
C1. Combined contingent liabilities	21	10	26	24	22	24	27	22	18	16	14
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

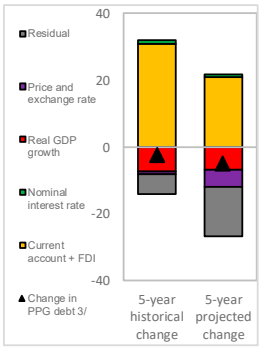
Sources: Country authorities; and staff estimates and projections.
1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
2/ Includes official and private transfers and FDI.

Figure 3. Central African Republic: Drivers of Debt Dynamics - Baseline Scenario

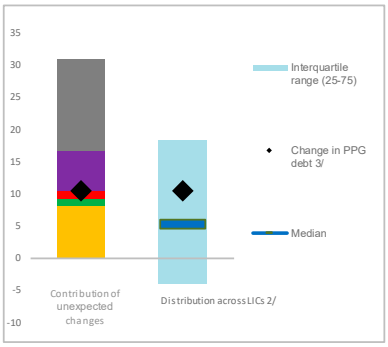
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

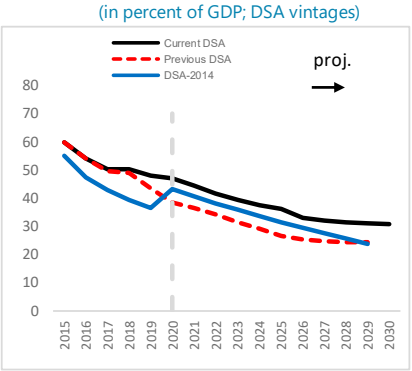


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

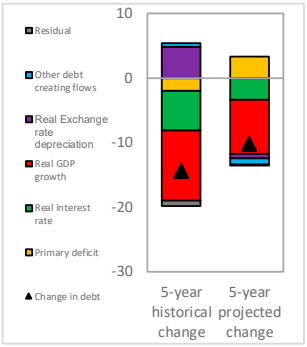


Public debt

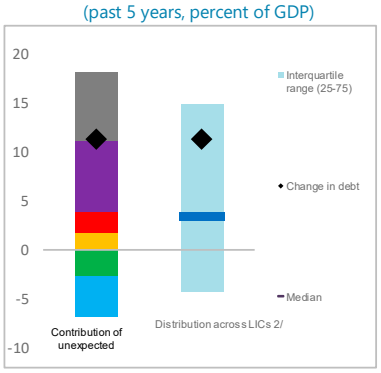
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



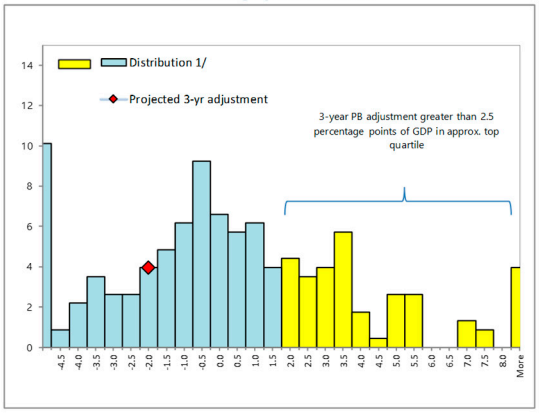
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

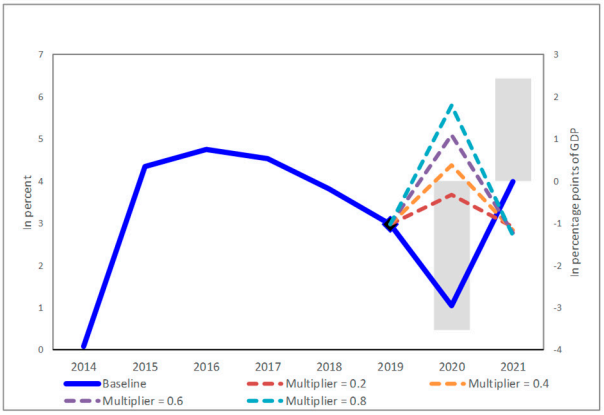
Figure 4. Central African Republic: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



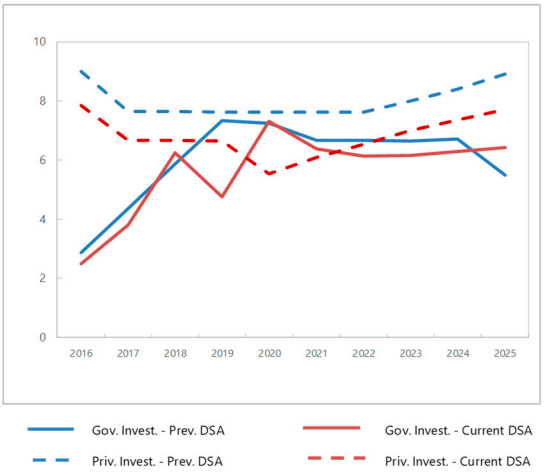
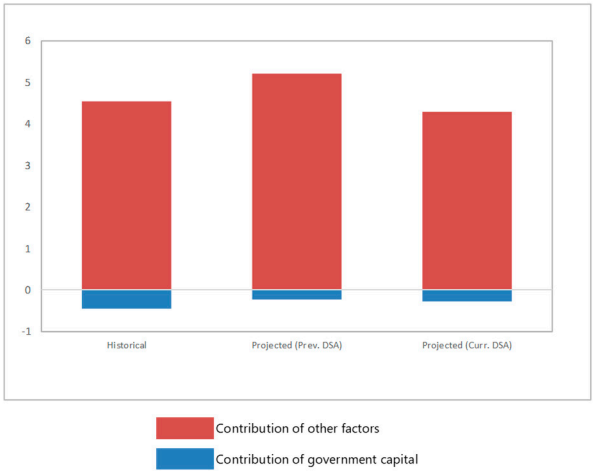
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)